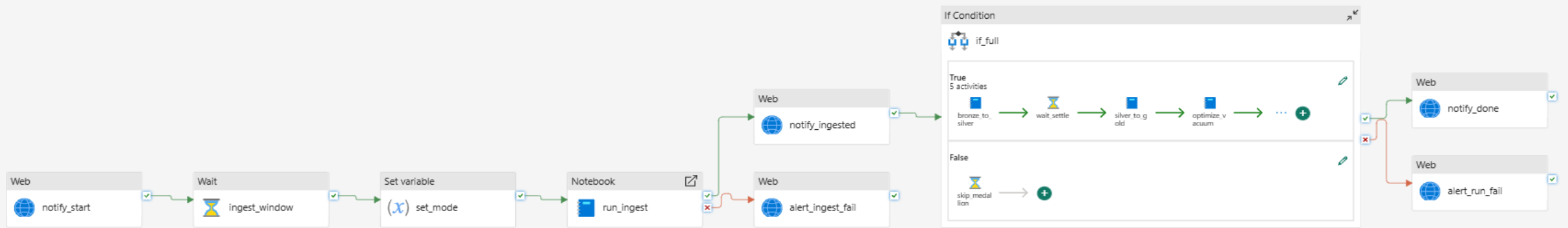


# Folder Overview

 ws\_mch\_iot > poc > historic analysis

|  | Name                                                                                    | Status | Type   | Task | Ow |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------|--------|------|----|
|  | analysis                                                                                |        | Folder | —    | —  |
|  | docs                                                                                    |        | Folder | —    | —  |
| <input type="checkbox"/>                                                          |  model | ...    | Folder | —    | —  |
|  | notebook                                                                                |        | Folder | —    | —  |
|  | pipelines                                                                               |        | Folder | —    | —  |
|  | report                                                                                  |        | Folder | —    | —  |
|  | storage                                                                                 |        | Folder | —    | —  |

# Pipeline Architecture:



## Schedules

**Every day** On

|                    |                               |                         |
|--------------------|-------------------------------|-------------------------|
| <b>Time of day</b> | <b>Last successful update</b> | <b>Next update</b>      |
| 12:00 AM           | Not available ⓘ               | 11 hour(s) 39 minute(s) |

|              |            |                                  |
|--------------|------------|----------------------------------|
| <b>Start</b> | <b>End</b> | <b>Time zone</b>                 |
| 06/21/2026   | 06/21/2027 | (UTC) Coordinated Universal Time |

**Job type**   **Modified By**  
Pipeline   Anuj Fabric ⓘ

**Parameters passed**  
None configured

⌵ Show less

ID: 67169086-5d5a-4de5-9ac2-c6ac1906c325 Run Edit

## Notebook Snippets:

```
# OPTIMIZE + V-Order compaction
spark.sql("OPTIMIZE delta.`Tables/brz/evt`")
print("appended", new.count(), "rows to brz.evt")
```

```
hwm = spark.sql("SELECT max(hwm_value) h FROM delta.`Tables/ctl/wm` WHERE layer='silver' AND source='brz.evt'").first()["h"]
bz = spark.read.format("delta").load("Tables/brz/evt")
inc = bz.where(F.col("event_ts") > F.expr(f"timestamp('{hwm}') - INTERVAL 24 HOURS")) if hwm else bz
```

```
agg = (ft.groupBy("machine_id", "date_sk")
      .agg(F.count("*").alias("events"),
           F.sum(F.col("is_alert").cast("int")).alias("alerts"),
           F.sum(F.col("is_anomaly").cast("int")).alias("anomalies"),
           F.sum(F.col("is_late").cast("int")).alias("late_events"),
           F.sum(~F.col("status").isin("IDLE", "MAINTENANCE", "OFFLINE")).cast("int").alias("running"),
           F.min("event_ts").alias("first_ts"), F.max("event_ts").alias("last_ts"))
      .withColumn("availability_pct", F.round(100.0 * F.col("running") / F.col("events"), 2))
      .withColumn("alert_rate_pct", F.round(100.0 * F.col("alerts") / F.col("events"), 2))
      .withColumn("anomaly_rate_pct", F.round(100.0 * F.col("anomalies") / F.col("events"), 2))
      .withColumn("late_pct", F.round(100.0 * F.col("late_events") / F.col("events"), 2))
      .withColumn("mtbf_hours", F.round(
          (F.col("last_ts").cast("long") - F.col("first_ts").cast("long")) / 3600.0
          / F.greatest(F.col("alerts") + F.col("anomalies"), F.lit(1), 3)))
      (agg.write.format("delta").mode("overwrite").option("overwriteSchema", "true").save("Tables/gld/agg_machine_daily"))
spark.sql("OPTIMIZE delta.`Tables/gld/ft_telemetry`")
spark.sql("OPTIMIZE delta.`Tables/gld/agg_machine_daily`")
```

## Lakehouse and Warehouse Overview:

### Explorer

Add lakehouses

- lh\_mch
  - Tables
    - dbo
    - brz
      - evt
    - ctl
    - gld
    - slv
      - dlq
      - evt
  - Files



### Explorer

Add lakehouses

- lh\_mch
  - Tables
    - dbo
    - brz
    - ctl
    - gld
    - slv
      - dlq
      - evt
  - Files



### lh\_mch

- Tables
  - dbo
  - brz
    - evt
  - ctl
  - gld
    - agg\_machin...
    - dim\_date
    - dim\_line
    - dim\_machine
    - dim\_plant
    - ft\_telemetry
  - slv
- Files

### Explorer

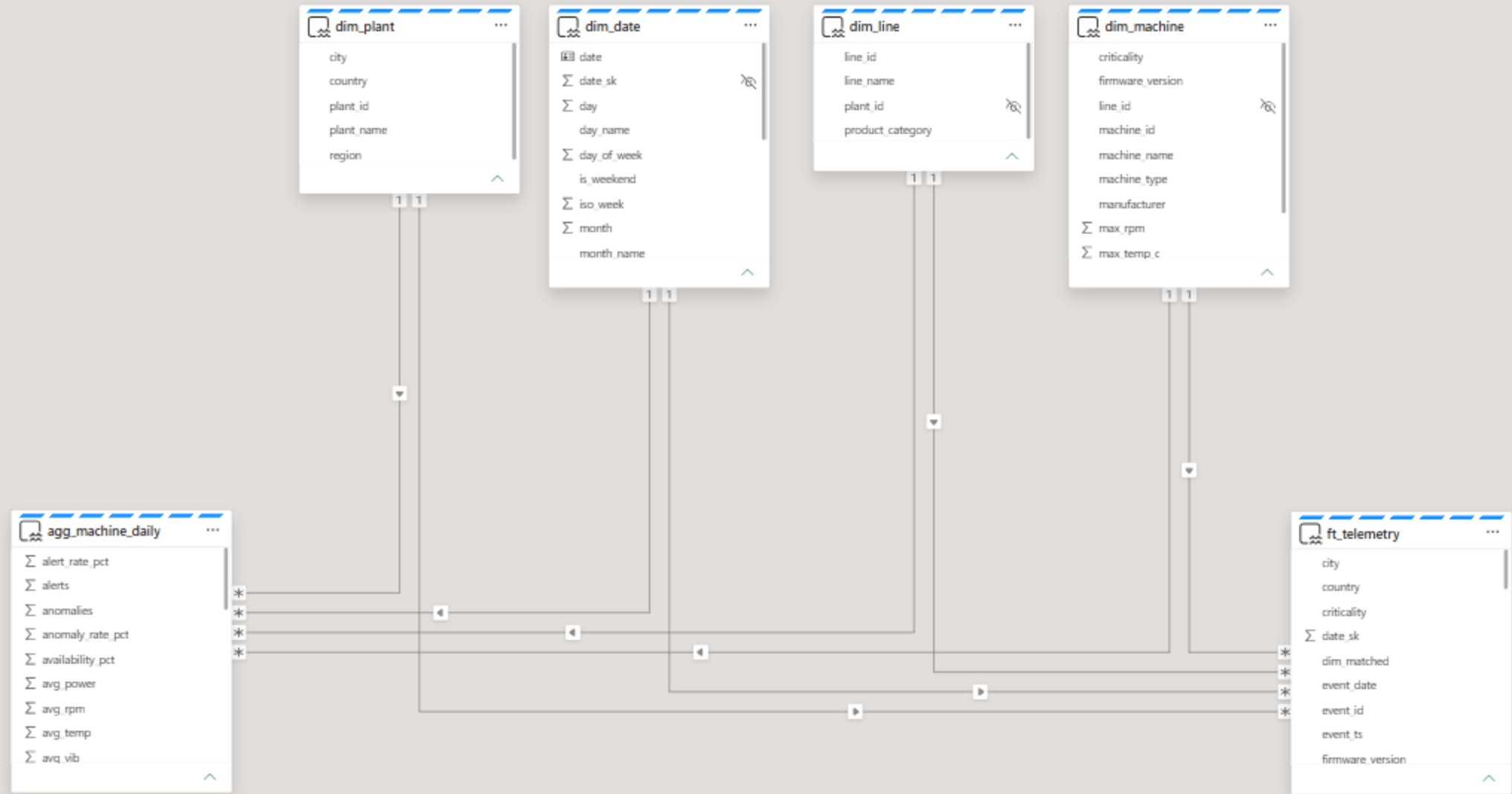
Add lakehouses

- lh\_mch
  - Tables
    - dbo
    - brz
    - ctl
      - freshness
      - wm
    - gld
    - slv
  - Files

### wh\_mch

- Schemas
  - dbo
  - dim
    - Tables
      - DIM\_LINE
      - DIM\_MACHINE
      - DIM\_PLANT
    - Views
    - Functions
    - Stored Procedures
  - INFORMATION\_SCHEMA
  - queryinsights
  - sys
  - Security

# Semantic Model:



# Dashboard :

Availability %

85.8%

Availability %

OEE % (proxy)

25.6%

OEE % (proxy)

Alert Rate %

58.2%

Alert Rate %

Anomaly Rate %

11.9%

Anomaly Rate %

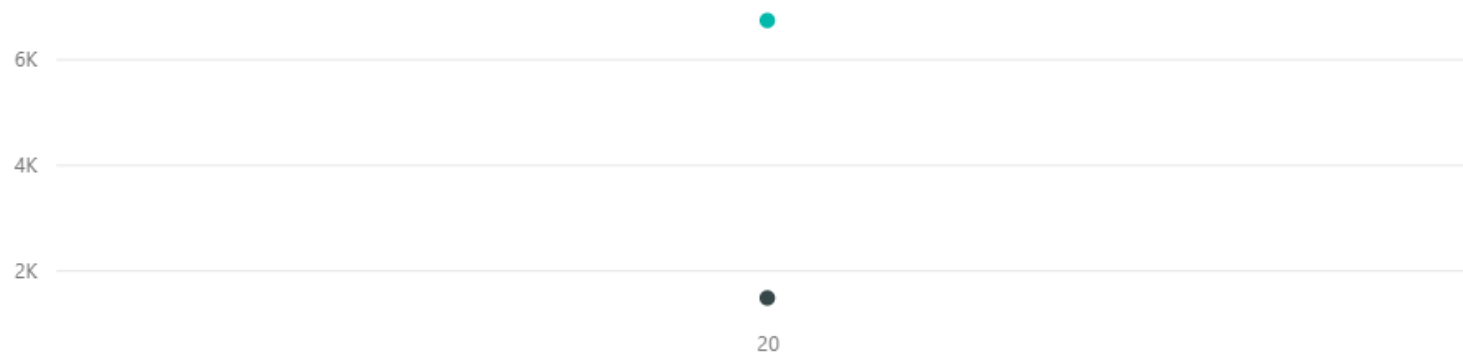
Machines

31

Machine Count

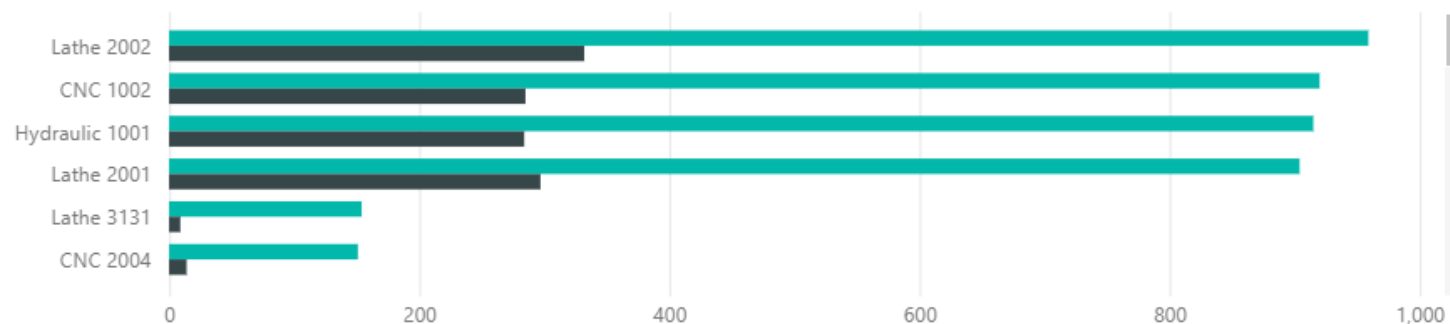
Alert & Anomaly Trend Over Time (drill: Year > Quarter > Month > Day)

● Alert Count ● Anomaly Count



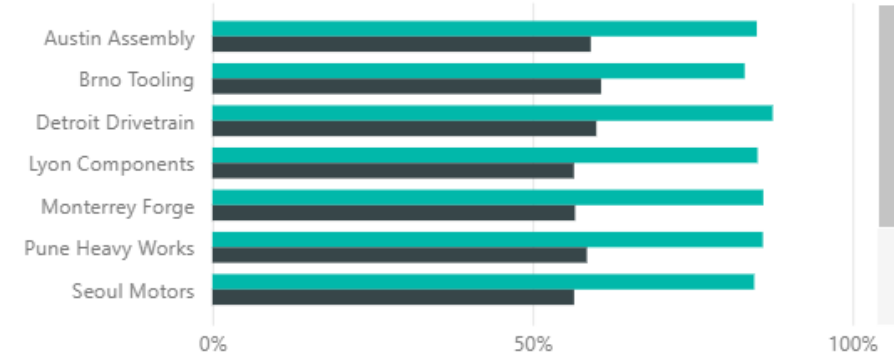
Top Offending Machines (Alerts + Anomalies)

● Alert Count ● Anomaly Count



Availability & Alert Rate by Plant

● Availability % ● Alert Rate %



Event Volume by Region

